Alan (Yulun) Wu

Data Scientist | ML Enthusiast | Specializing in Predictive Analytics & Data Solutions

University of Pennsylvania

Hello! I'm Alan Wu, a graduate student in Data Science at the University of Pennsylvania and a Mathematics alumnus from the University of Nottingham. My passion lies in transforming complex data into actionable insights using advanced analytics, statistical modeling, and scalable data engineering tools. I'm dedicated to harnessing data-driven strategies that spark innovation and deliver tangible business impact. When I'm not coding or analyzing datasets, you can find me exploring the latest data science research or refining my visualization skills to bring stories to life through data. Welcome to my space – feel free to explore my projects and connect!

Download Résumé

Education

Master of Science in Data Science

University of Pennsylvania
August 2024 – May 2026

- Overall GPA: 4.0/4.0
- Pursuing advanced studies in data science, applied machine learning, and AI.

Bachelor of Science (Honours) in Mathematics with Applied Mathematics

University of Nottingham
September 2020 – July 2024

- Overall GPA: 3.9/4.0 | First Class | Provost's Scholarship (06/2021)
- Specialized in probability theory, statistical methods, and mathematical modeling.

Experience

Technical Lead

Wharton Analytics Fellows
February 2025 – April 2025

- Collaborating within the Wharton AI & Analytics Accelerator Team to develop an AI-driven tool that predicts sales trends for Spencer's Gifts, focusing on t-shirts and other trendy items.
- Analyzing historical and current sales data to forecast changes in sales magnitude, timing, and revenue impact.
- Preparing weekly reports with actionable insights for the client's Planning Team to support strategic decision-making.

Data Analyst Intern

Novo Nordisk (China) Pharmaceuticals Co., Ltd.
June 2023 – September 2023

- Collaborated with Microsoft China on a company-wide SQL database migration project regarding to customer representative behavior, consolidating five databases into one. Reduced query runtime by 30% through SQL optimization, increasing database migration efficiency.
- Leveraged Python and statistical techniques to process and analyze historical sales promotion data for pharmaceuticals. Generated predictive insights that improved promotional budget allocation accuracy by 20%, contributing to $100,000 in cost savings.

Data Analyst Intern

Central Institute of Mental Health
January 2023 – May 2023

- Explored neurophysiological activity under stress conditions using a neuropsychological paradigm and combining functional MRI and heart rate signal (in collaboration with the Max Planck Institute of Psychiatry).
- Analyzed fMRI-BOLD and photoplethysmogram signals respectively utilizing generalized linear models (GLM) and wabp algorithm in MATLAB and found brain regions' interaction with the autonomic nervous system under stress.
- Collaborated with a multidisciplinary team of researchers to prepare and deliver a comprehensive presentation at an academic seminar at Heidelberg University, engaging over 20 scholars and researchers in discussions about findings.

Skills

Software Skills

- PYTHON
- R
- SQL
- MATLAB
- C/C++
- POWERBI
- TABLEAU

Analytic Skills

- STATISTICAL MODELING
- A/B TESTING
- CAUSAL INFERENCE
- DATA CLEANING
- DATA MINING
- DATA VISUALIZATION
- DATA ENGINEERING

Projects

News Source Classification

University of Pennsylvania · December 2024

- Developed binary classification models to predict news headlines' sources (Fox News vs. NBC News), leveraging advanced NLP techniques to reveal stylistic and vocabulary differences.
- Web-scraped and processed 3,800+ headlines using Python, transforming text data via TF-IDF and Word2Vec vectorization, and enriching features through bigram/trigram extraction.
- Built and tuned a Bidirectional LSTM model (80% accuracy), outperforming a baseline Logistic Regression (66%); mitigated overfitting through extensive hyperparameter optimization.

Credit Card Fraud Prediction

University of Pennsylvania · December 2024

- Led a credit card fraud prediction project using Python and SQL, analyzing over 50,000 transaction records to identify key patterns and factors contributing to fraudulent activities, such as transaction amount, location anomalies, and user behavior.
- Utilized SMOTE and ADASYN methods to address data imbalance, increasing sample size for robust model training. Enhanced feature extraction by leveraging Long Short-Term Memory (LSTM) networks for detecting temporal patterns.
- Built and evaluated classification models including Logistic Regression, Random Forest, and XGBoost, achieving a 90% prediction accuracy with XGBoost. Explained model predictions using SHAP value charts, enabling stakeholders to understand key drivers of fraud.

Customer Churn Analysis

· September 2024

- Led a comprehensive customer churn analysis using Python, analyzing data from over 7,000 customer records to identify key factors such as contract length, monthly charges and customer service response time influencing churn rates.
- Applied the SMOTE method to increase sample size and developed multiple classification models (Logistic Regression, Random Forest, XGBoost) to predict potential customer churn. Achieved an accuracy of 0.9 using XGBoost and created SHAP value charts to explain the model.
- Designed and implemented A/B testing to validate data-driven strategies for reducing customer churn, comparing pre-and post-intervention churn rates to measure the effectiveness of the campaign.

Dimension Reduction Analysis

· May 2023

- Implemented PCA and CAE techniques for data compression and decompression on 48 videos with different initial conditions related to Reduced Order Modeling & Video Prediction.
- Compared the reconstruction accuracy of PCA and CAE against various dimensions of the reduced space and found better accuracy for larger dimensions using CAE and for smaller dimensions using PCA.
- Developed a sequence-to-sequence predictive model (LSTM) for four timesteps in the reduced space and achieved a 92% accuracy rate in decoding predicted results in the full space.

© 2025 Me. This work is licensed under CC BY NC ND 4.0
Published with Hugo Blox Builder — the free, open source website builder that empowers creators.